Supplementary Step 11 Visualization Report

Overview

Metric	Value
Included participant-sessions	36
Included questions	30
Included trials	1075
Primary robust SE type	Two-way cluster (participant + question)
Primary continuous p-value	0.303503
Binary benchmark p-value	0.209749
Quadratic linear-term p-value	0.188168
Quadratic squared-term p-value	0.218468
Threshold-strength p-value	0.504510
Back ROI benchmark p-value	0.404042
Fixed-window benchmark p-value	0.271618
No-override p-value	0.481741
Binary cutoff used	0.50
Questions below / above cutoff	15 / 15

Interpretation note. In the participant-fixed-effects regression using the continuous abstractness score, the association between abstractness level and left frontal ROI HbO response was not statistically significant (beta=1.311e-08, SE=1.251e-08, p=0.303503, 95% CI [-1.249e-08, 3.870e-08]). This suggests that replacing the binary label with a continuous abstractness measure does not by itself yield robust evidence for a frontal abstractness effect in the current dataset. The binary benchmark remained non-significant (p=0.209749), the fixed-window continuous benchmark remained non-significant (p=0.271618), and the no-override continuous model remained non-significant (p=0.481741).

Correlation note. Score-diagnostic correlations were: Total word count: r=0.040, p=0.832348; Sentence count: r=-0.011, p=0.953914; Average sentence length: r=0.035, p=0.855832; ENEM item correctness: r=0.400, p=0.028305; Mean response time: r=0.010, p=0.956933.

Core Step 11 Figures

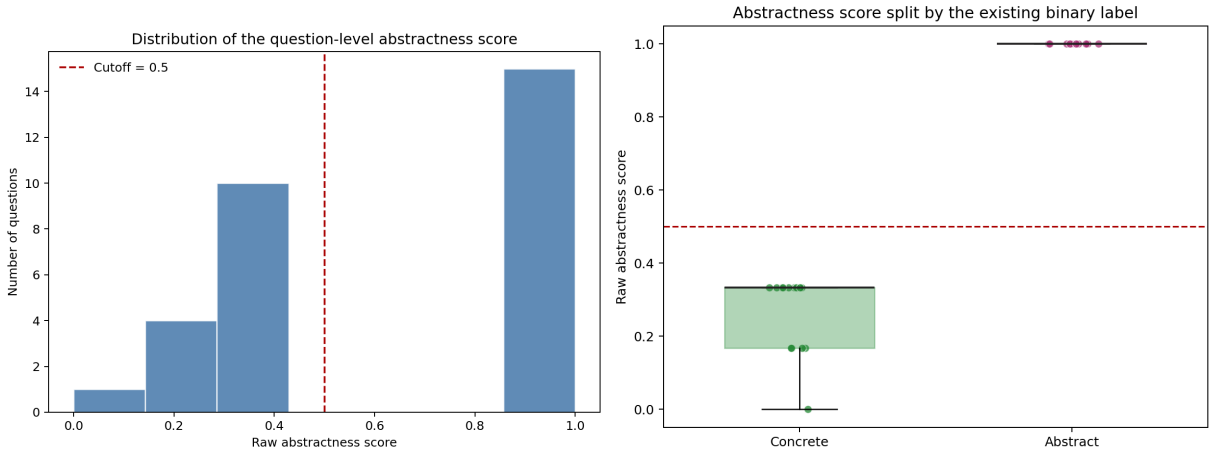

Figure 1: Distribution of the raw abstractness score and its split by the existing binary label.

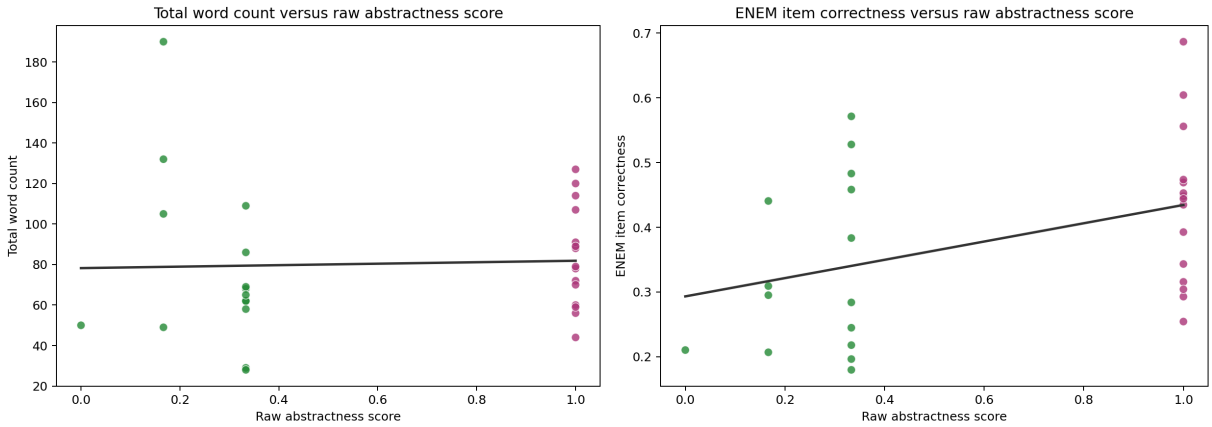

Figure 2: Abstractness score against total word count and ENEM item correctness.

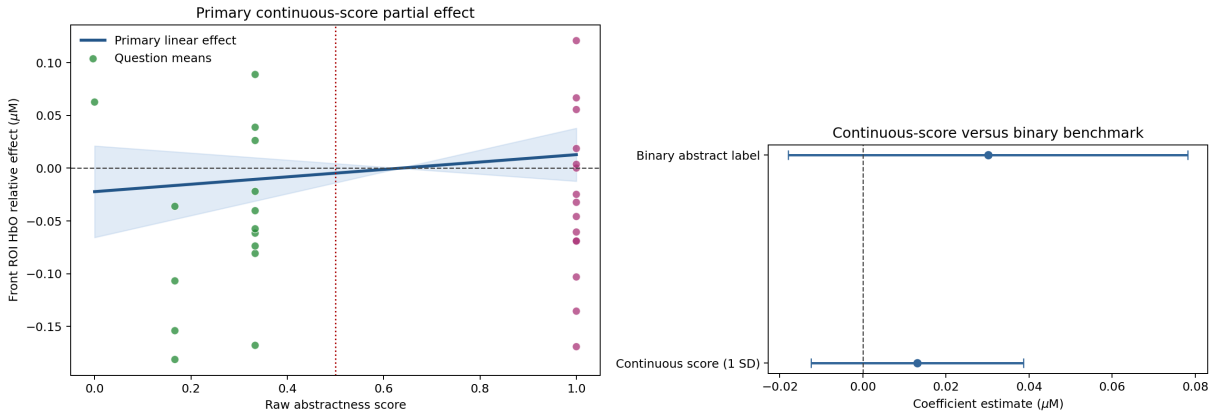

Figure 3: Primary continuous-score partial effect and the comparison against the binary benchmark.

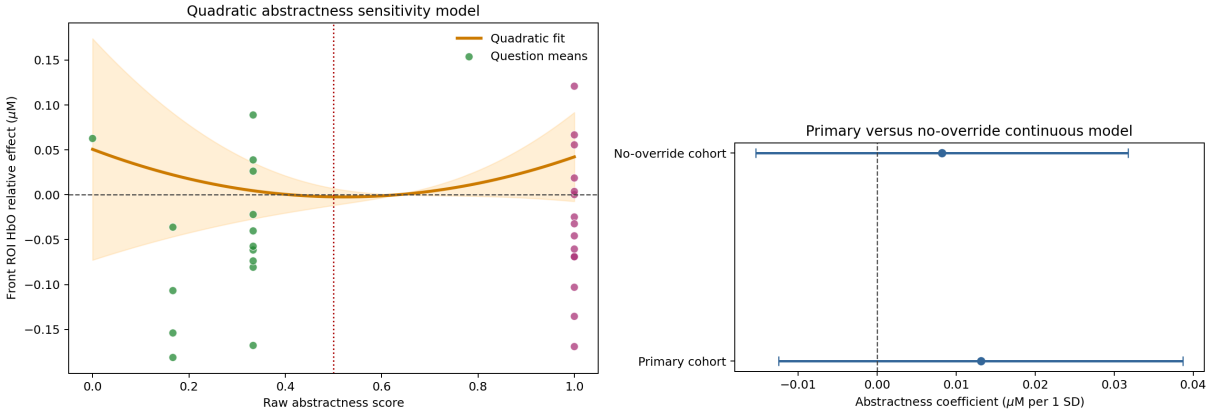

Figure 4: Quadratic sensitivity visualization and the no-override comparison.

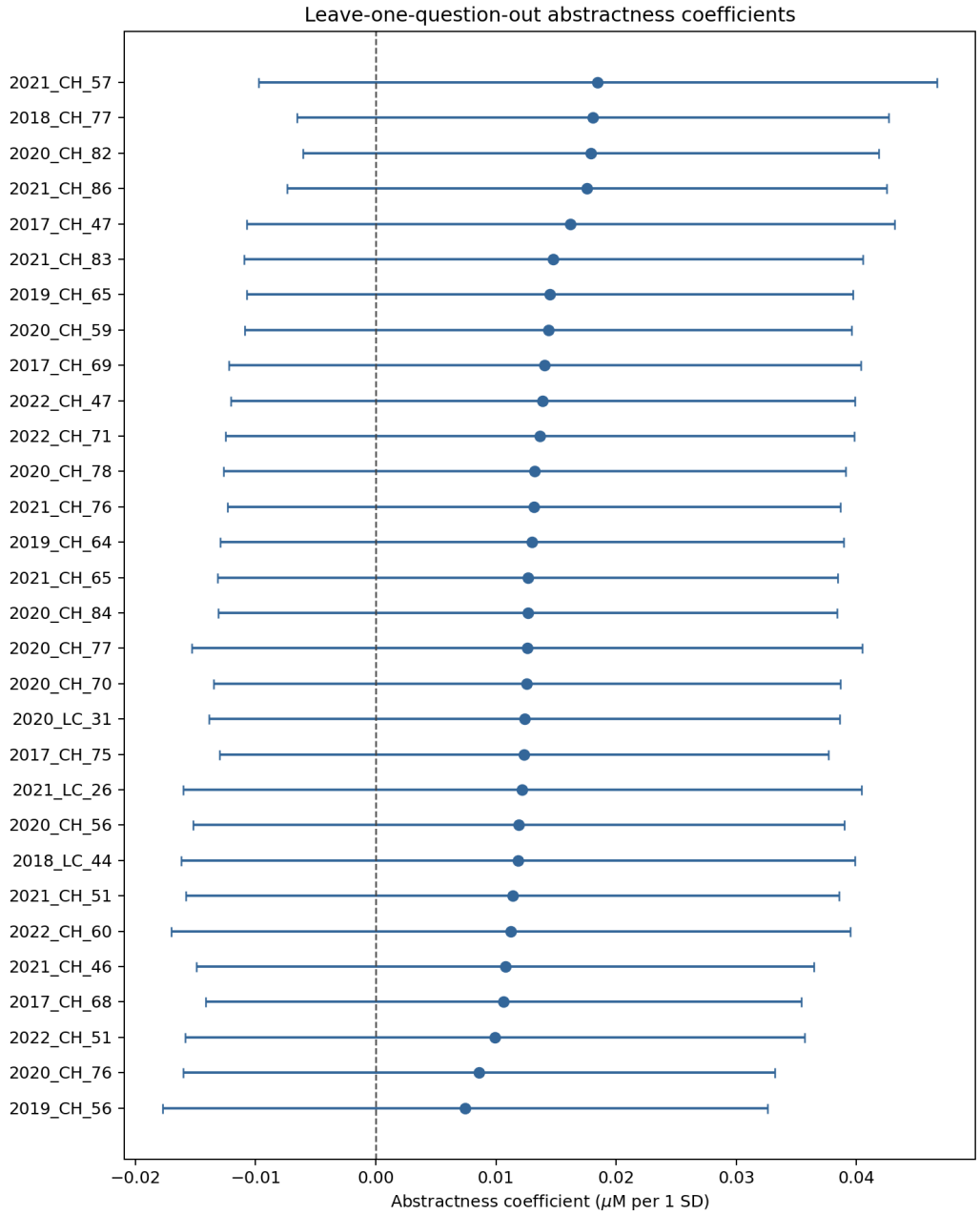

Figure 5: Leave-one-question-out abstractness coefficients from the main Step 11 run.

Additional Step 11 Diagnostics

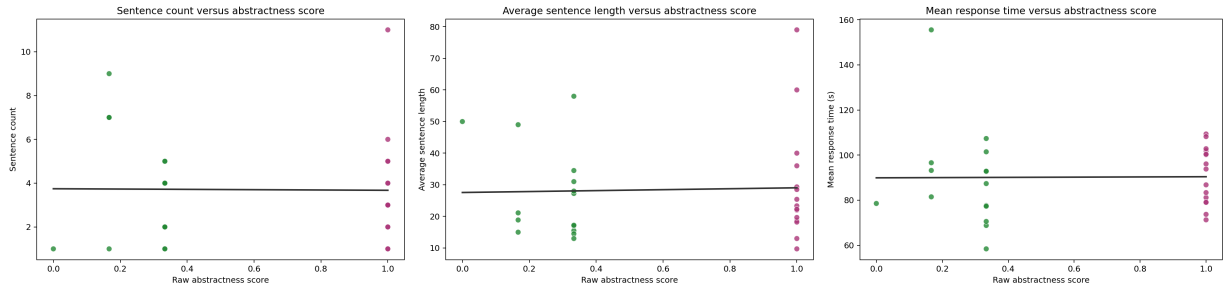

Figure 6: Additional score-diagnostic scatter plots for sentence count, sentence length, and mean response time.

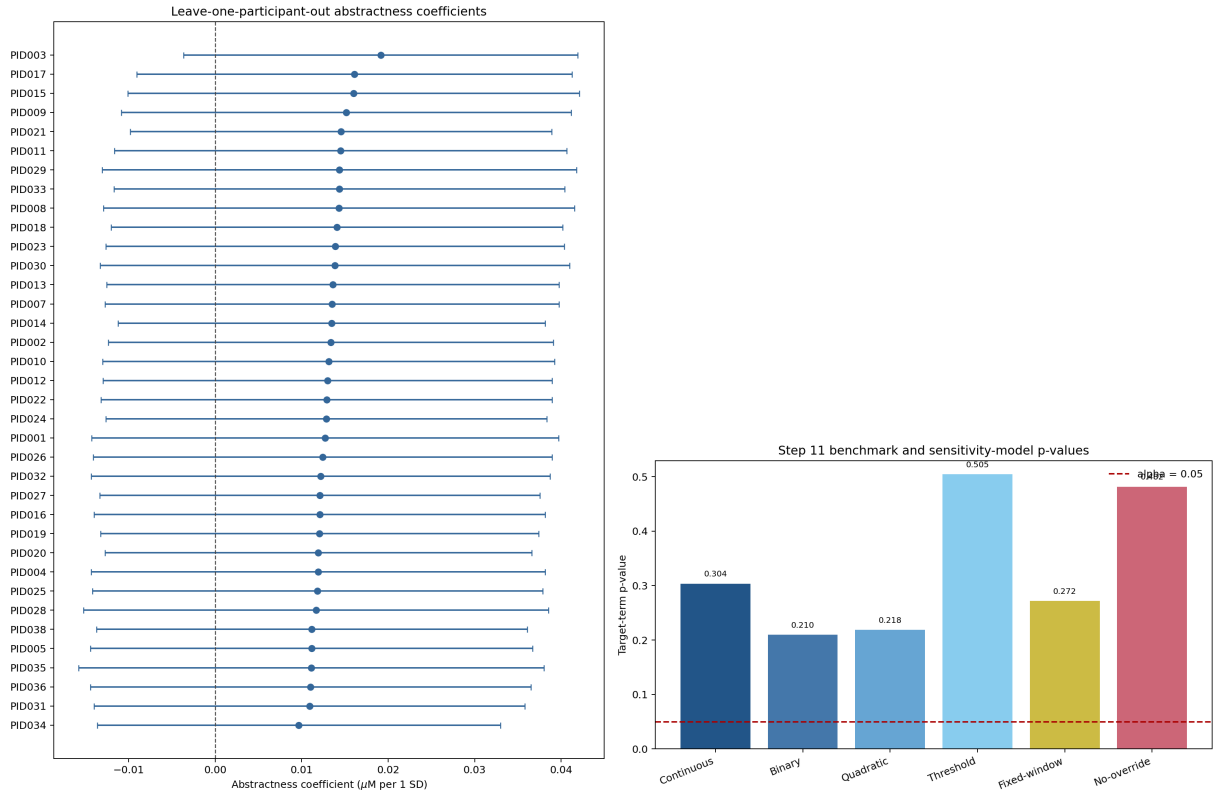

Figure 7: Leave-one-participant-out abstractness coefficients and the p-value summary across Step 11 benchmark/sensitivity models.

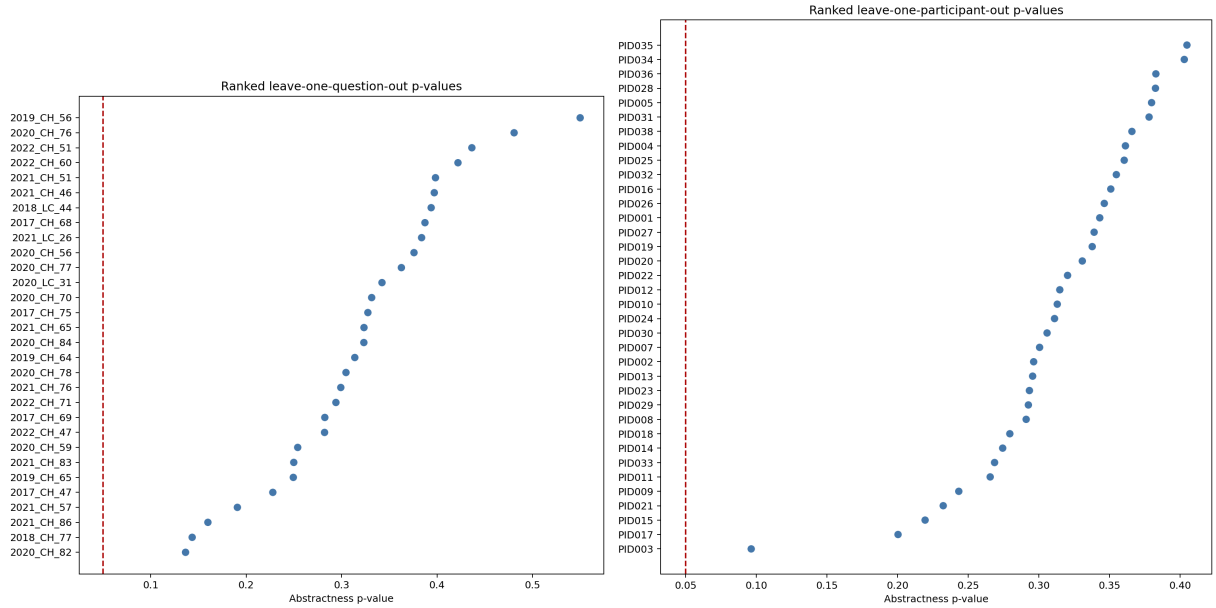

Figure 8: Ranked p-values from the leave-one-question-out and leave-one-participant-out analyses. The red line marks alpha=0.05.